

5(a)	<i>any two from:</i> (H because) it is closed / blood is (transported) in blood vessels (and heart) / blood cannot leave the system ; (because) it, passes through the heart twice (in one complete circuit) / is double circulation ; (because) it, shows pulmonary and systemic, circulations / circuits ; <i>If letter J given as the answer, allow one mark for 'closed'.</i> R (letter) G	2
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Question	Answer	Marks
5(b)	any four from: (at arteriole end of capillary) high, hydrostatic / blood, pressure ; (pressure / ultra) filtration (of blood) ; plasma / water and solutes / fluid, leave / AW, blood / capillaries ; R diffusion glucose / amino acids / ions, (filtered) out ; (large) plasma proteins, remain / not filtered out ; ref. to endothelial pores / fenestrations / fenestrae or pores / gaps / spaces, in, capillary wall / endothelium / between (endothelial) cells ; AVP ; e.g. hydrostatic pressure of blood is greater than hydrostatic pressure of tissue fluid (at arteriole end of capillary)	4

Question	Answer	Marks
5(c)	<i>allow ecf if symplast and apoplast pathway are not named but are incorrectly assigned</i> any five from: <i>symplast pathway</i> water enters, root hair cell / epidermal cell, by osmosis (across cell surface membrane) ; water movement through cytoplasm (of cells) ; <i>ref. to vacuolar pathway ;</i> A described e.g. crosses tonoplast of vacuole(s) enters / leaves, vacuole(s) by osmosis water movement to adjacent cells via plasmodesmata ; AW <i>apoplast pathway</i> cell wall route ; further detail ; e.g. between, cellulose fibres / spaces between cells / intercellular spaces <i>at endodermis</i> enters cells when endodermis reached / moves into symplast pathway at endodermis ; <i>ref. to blocked by, suberin / Casparian strip (at endodermis) ;</i> AVP ; e.g. correct sequence of, cell / tissue, pathway <i>three of the four of</i> epidermis / epidermal cell / root hair cell to, cortex / cortical cells / parenchyma cells to, pericycle to, endodermis / endodermal cell	5